

Internship Report



**DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY GOA**

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Comprehensive Analysis of Binary Divider Architectures

Introduction to Binary Division

Binary division is one of the most complex and resource-intensive arithmetic operations to implement in digital hardware. Unlike addition or multiplication, division is inherently sequential, requiring the hardware to evaluate intermediate remainders to generate quotient bits.

This report explores and analyzes five distinct hardware division architectures, broadly categorized into two families:

1. **Digit-Recurrence Algorithms:** These methods compute the quotient bit-by-bit using subtraction and shifting operations. They include **Restoring**, **Non-Restoring**, and **SRT** division. These architectures are generally slower but require minimal hardware area.
2. **Multiplicative Algorithms:** These high-speed methods approximate the reciprocal of the divisor and iteratively refine the result using multiplication. They include the **Newton-Raphson** and **Goldschmidt** algorithms. These architectures converge quadratically, making them incredibly fast, but they rely on massive parallel multiplier trees and Look-Up Tables (LUTs).

The following chapters break down the mathematical foundation, hardware architecture, and circuit flow of each divider variant, concluding with a comparative analysis of their area-delay trade-offs for an 8-bit datapath.

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1 Restoring Divider

1.1 Algorithm and Working Principle

The restoring division algorithm is a sequential, cycle-by-cycle method that calculates a division result one bit at a time, mimicking how we perform long division on paper. The fundamental loop consists of three distinct steps: shift, subtract, and check.

Traditionally, the basic idea relies on shifting a large, combined register containing the remainder and the dividend. As the dividend bits shift left, they slowly feed into the remainder space. The divisor is then subtracted from this new partial remainder.

- If the subtraction yields a negative result (meaning the divisor was too big), the hardware "restores" the remainder back to what it was before the subtraction, and sets the current quotient bit to 0.
- If the subtraction yields a positive result or zero, it keeps the newly subtracted value as the new remainder and sets the quotient bit to 1.

Our Hardware Implementation (The Concatenation Method): While the textbook theory often implies using a massive unified shift register or two separate active shifters to move the remainder and quotient, our hardware implementation uses a much more efficient approach: wire concatenation.

Instead of a second shifter for the remainder, we dynamically build the shifted partial remainder directly on the wires. During each cycle, the hardware automatically concatenates the lower 7 bits of the current Remainder register with the Most Significant Bit (MSB) that is being pushed out of the Quotient (Qu) register. This concatenated value is then instantly fed into the subtractor for the math operation.

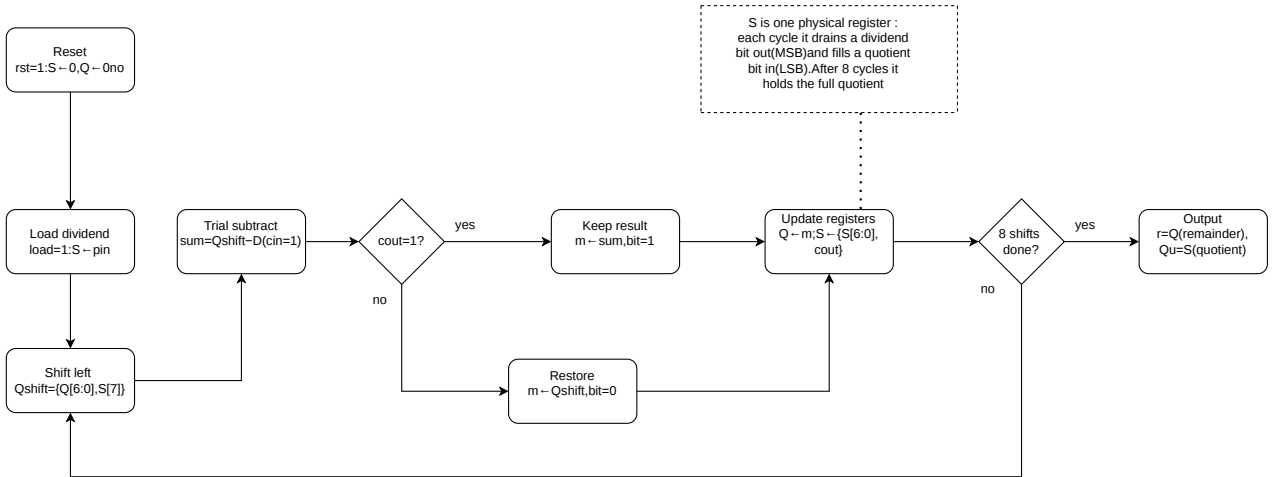


Figure 1: Logical Flowchart of the Restoring Division Algorithm

1.2 Hardware Architecture and Components

To execute the restoring algorithm, the physical datapath is constructed using the following core digital logic blocks:

- **The Shifter:** This component initially holds the 8-bit Dividend. On every clock cycle, it shifts left, passing its Most Significant Bit (MSB) out to the concatenation wire to form the partial remainder, while pulling in the newly calculated quotient bit at its Least Significant Bit (LSB).
- **The Remainder Register (D Flip-Flops):** An array of 8 individual D flip-flops stores the current partial remainder. It updates synchronously with the clock based on the routing decision made by the MUX.
- **The Subtractor (8-Bit Full Adder):** The subtraction step is executed by a standard 8-bit full adder. By feeding the Divisor into input B and tying the carry-in (**cin**) to a permanent high signal (1), the adder effectively performs two's complement subtraction on the concatenated partial remainder.
- **The Decision Logic (Multiplexer):** A standard MUX acts as the critical control block. The carry-out (**cout**) bit from the Adder directly drives the select line of the MUX.
 - If **cout** is 1 (indicating a valid, positive subtraction), the MUX routes the Adder's new output into the D flip-flops.
 - If **cout** is 0 (indicating a negative overflow), the MUX safely routes the original un-subtracted concatenated wire value back to the flip-flops, physically executing the "restore" action without wasting an extra clock cycle.

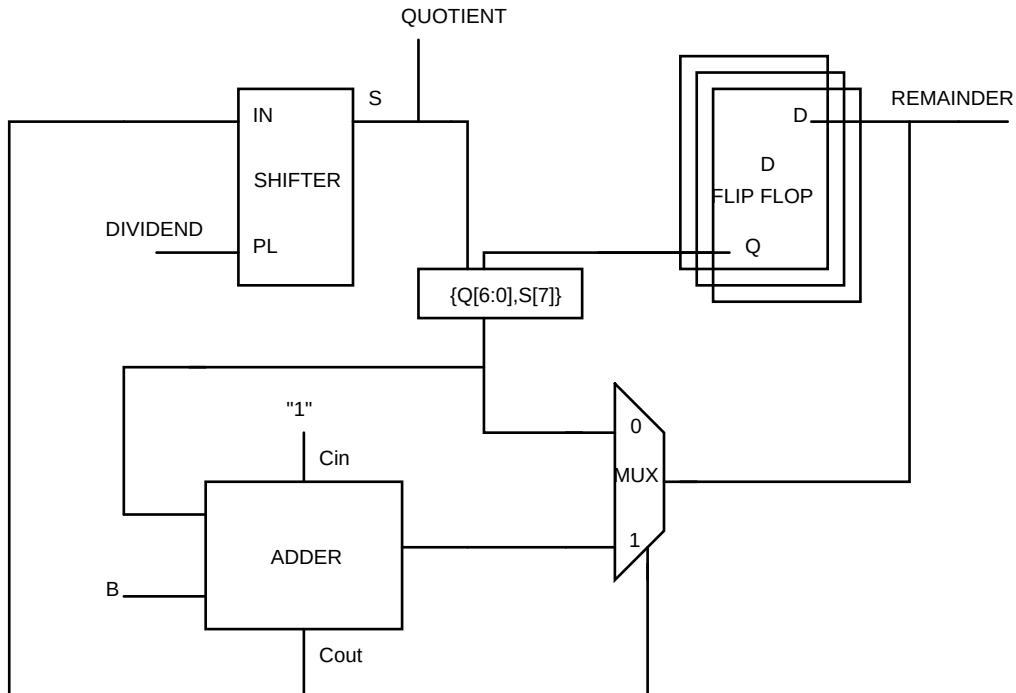
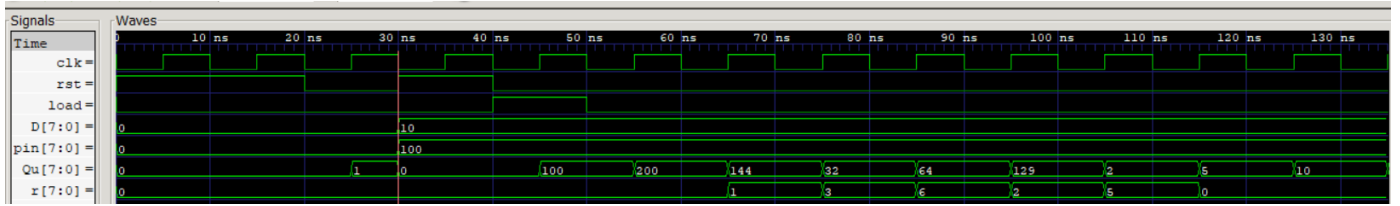


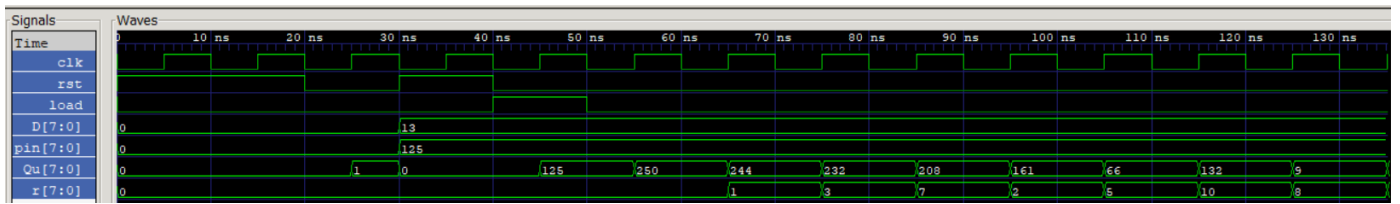
Figure 2: Main Circuit Schematic of the Restoring Divider

1.3 Simulation Results

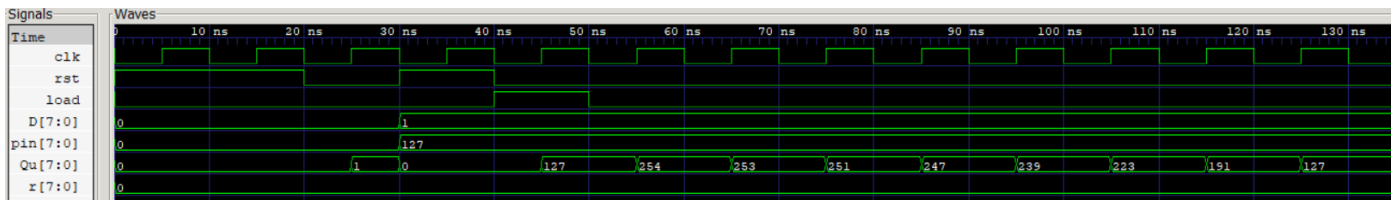
The architecture was rigorously tested using four specific edge cases. The state transitions across all simulations matched the theoretical math exactly.



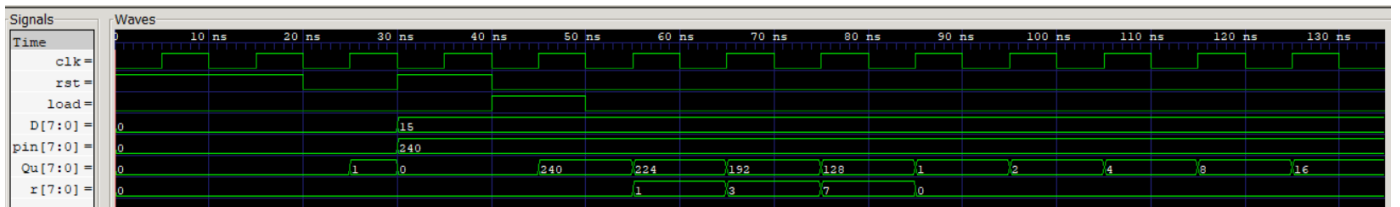
(a) Test Case 1: 100 / 10 (Q=10, R=0)



(b) Test Case 2: 125 / 13 (Q=9, R=8)



(c) Test Case 3: 127 / 1 (Q=127, R=0)



(d) Test Case 4: 240 / 15 (Q=16, R=0)

Figure 3: Timing Simulation Results for the Restoring Divider.

1.4 Cycle-by-Cycle Execution Trace (100 / 10)

To demonstrate how this hardware seamlessly manages the math, we trace the division of 100 by 10 below. The system begins with the Divisor permanently set to 10. The Quotient register (Qu) holds the initial dividend of 100 (binary 01100100). The Remainder register (r) starts at 0.

- **Cycle 1:**
 - **Shift Action:** Qu shifts left. Its MSB (0) moves into r. The empty LSB slot at the end of Qu waits for a new bit, temporarily making Qu read as 200 (binary 11001000). The concatenated r is now 0.
 - **Math & Decision:** $0 - 10$ is negative. The MUX restores r to 0. A 0 is fed into the waiting LSB of Qu.
 - **End of Cycle State:** $r = 0$, $Qu = 200$.
- **Cycle 2:**
 - **Shift Action:** Qu shifts left. Its MSB (1) moves into r. Qu temporarily becomes 144 (10010000). The concatenated r becomes 1.
 - **Math & Decision:** $1 - 10$ is negative. The MUX restores r to 1. A 0 is fed into the LSB of Qu.
 - **End of Cycle State:** $r = 1$, $Qu = 144$.
- **Cycle 3:**
 - **Shift Action:** Qu shifts left. Its MSB (1) moves into r. Qu temporarily becomes 32 (00100000). The concatenated r becomes 3.
 - **Math & Decision:** $3 - 10$ is negative. The MUX restores r to 3. A 0 is fed into the LSB of Qu.
 - **End of Cycle State:** $r = 3$, $Qu = 32$.
- **Cycle 4:**
 - **Shift Action:** Qu shifts left. Its MSB (0) moves into r. Qu temporarily becomes 64 (01000000). The concatenated r becomes 6.
 - **Math & Decision:** $6 - 10$ is negative. The MUX restores r to 6. A 0 is fed into the LSB of Qu.
 - **End of Cycle State:** $r = 6$, $Qu = 64$.
- **Cycle 5:**
 - **Shift Action:** Qu shifts left. Its MSB (0) moves into r. Qu temporarily becomes 128 (10000000). The concatenated r becomes 12.
 - **Math & Decision:** $12 - 10 = 2$ (Positive). The MUX keeps the new subtraction result. r becomes 2. A 1 is fed into the LSB of Qu.
 - **End of Cycle State:** $r = 2$, $Qu = 129$.
- **Cycle 6:**

- **Shift Action:** Qu shifts left. Its MSB (1) moves into r. Qu temporarily becomes 2 (00000010). The concatenated r becomes 5.
- **Math & Decision:** 5 - 10 is negative. The MUX restores r to 5. A 0 is fed into the LSB of Qu.
- **End of Cycle State:** r = 5, Qu = 2.
- **Cycle 7:**
 - **Shift Action:** Qu shifts left. Its MSB (0) moves into r. Qu temporarily becomes 4 (00000100). The concatenated r becomes 10.
 - **Math & Decision:** 10 - 10 = 0 (Treated as success). The MUX keeps the subtraction result. r becomes 0. A 1 is fed into the LSB of Qu.
 - **End of Cycle State:** r = 0, Qu = 5.
- **Cycle 8:**
 - **Shift Action:** Qu shifts left. Its MSB (0) moves into r. Qu temporarily becomes 10 (00001010). The concatenated r becomes 0.
 - **Math & Decision:** 0 - 10 is negative. The MUX restores r to 0. A 0 is fed into the LSB of Qu.
 - **Final State:** The division concludes perfectly with Remainder = 0, Quotient = 10.

1.5 Performance Evaluation

- **Advantages:** The main advantage of the restoring divider is its logic and hardware simplicity. By utilizing only basic shift registers, a single adder, and a standard MUX, it maintains a highly area-efficient footprint on the chip.
- **Disadvantages:** The overall execution speed is slow. Because a subtraction and a potential restoring action must both safely resolve within a single clock cycle, this architecture suffers from a longer critical path delay compared to non-restoring methods.

2 Non-Restoring Divider

2.1 Algorithm and Working Principle

The non-restoring division algorithm is an iterative subtractive process that computes a quotient and remainder by shifting and performing Two's Complement arithmetic. Unlike restoring division, it does not waste a clock cycle correcting a negative partial remainder during the calculation loops. Instead, it temporarily leaves the remainder in a negative state (non-restored) and compensates in the subsequent clock cycle by dynamically switching the arithmetic operation from subtraction to addition.

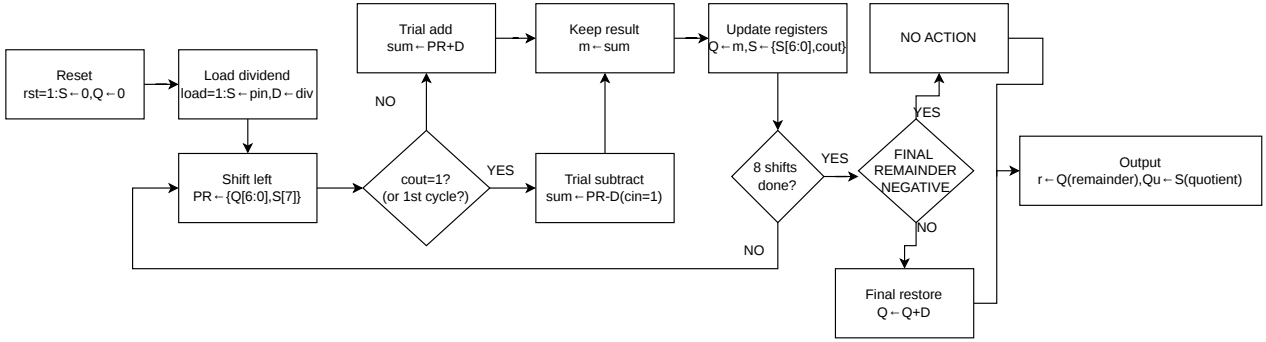


Figure 4: Logical Flowchart of the Non-Restoring Division Algorithm

The core control relies entirely on the sign of the current remainder:

- **If the remainder is Positive (or 0):** The system shifts the bits left and subtracts the divisor.
- **If the remainder is Negative:** The system shifts the bits left and adds the divisor.

The circuit operates continuously across 8 clock cycles. Because the algorithm allows negative intermediate values, the final cycle can occasionally finish with a negative remainder, which is resolved by a standalone combinatorial correction step.

2.2 Data Constraints and Bit-Widths

To maintain mathematical stability during Two's Complement execution, the core datapath relies on strict bit-width boundaries:

- **Dividend (8-Bit):** The circuit accepts a full 8-bit unsigned dividend (up to a decimal value of 255), loaded directly into the quotient tracking register.
- **Divisor (7-Bit Limit):** The divisor input is restricted to 7 bits (maximum value of 127) with the MSB padded to 0. This constraint prevents the magnitude of the shifted partial remainder from overflowing into the 8th bit, ensuring the ALU never mistakes a large positive magnitude for a negative sign bit.

2.3 Hardware Architecture and Components

The physical datapath avoids bulky external control blocks, implementing the core math using structured components detailed below alongside their respective schematics.

2.3.1 1. The Main Datapath

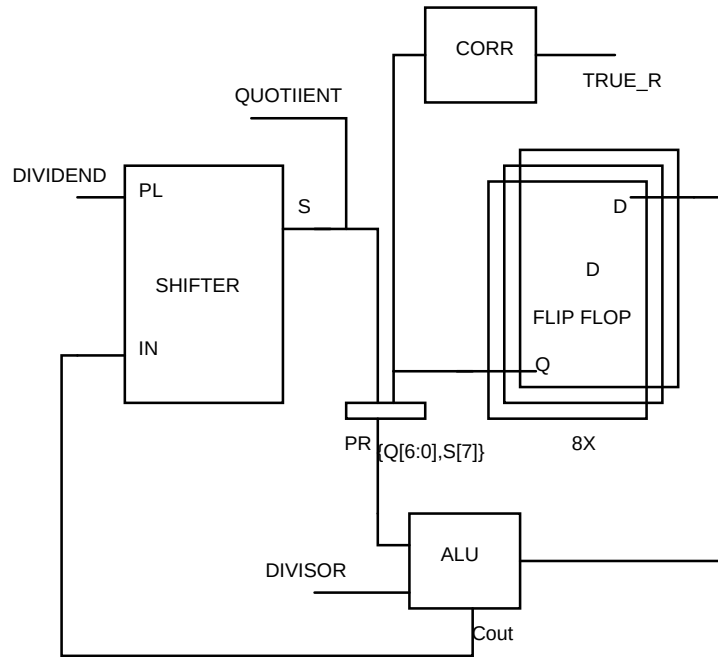


Figure 5: Main Circuit Schematic of the Non-Restoring Datapath

- **The Shifter:** Holds the dividend, shifting left every cycle to feed its MSB into the arithmetic path while capturing the incoming quotient bit.
- **The Concatenation Block:** Dynamically constructs the partial remainder directly on the wires by combining the lower 7 bits of the remainder register with the shifter's outgoing MSB ($\{Q[6:0], S[7]\}$).
- **The Remainder Register:** An 8-bit D flip-flop array storing the intermediate remainder state synchronously.

2.3.2 2. The ALU and Decision Logic

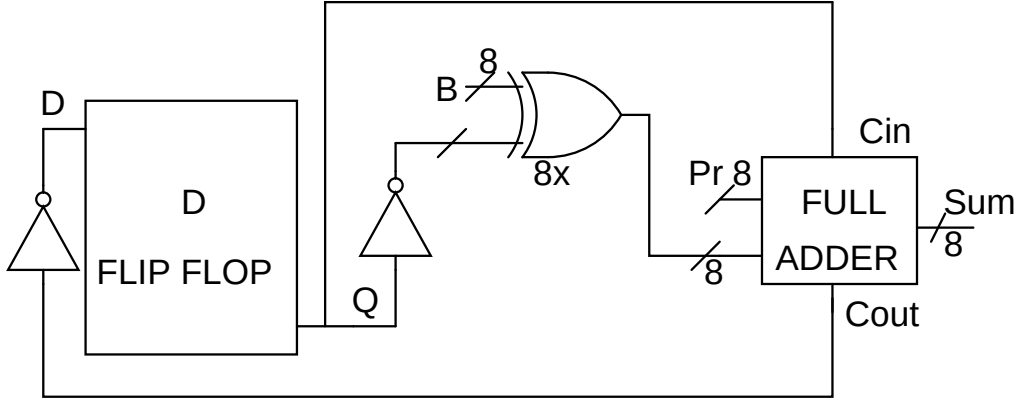


Figure 6: ALU Schematic Featuring Conditional XOR Inversion and Carry Routing

The ALU handles on-the-fly arithmetic switching via an XOR-based inverter configuration:

- **State Memory and XOR Array:** A flip-flop captures the inverted carry-out state, driving an 8-bit XOR gate array alongside the divisor to conditionally invert it for subtraction or pass it normally for addition.
- **Carry Routing (Cin):** The control line feeds directly into the adder's carry-in, executing Two's complement subtraction ($A + \sim B + 1$) when positive or addition ($A + B$) when negative.

2.3.3 3. The Correction Block (CORR)

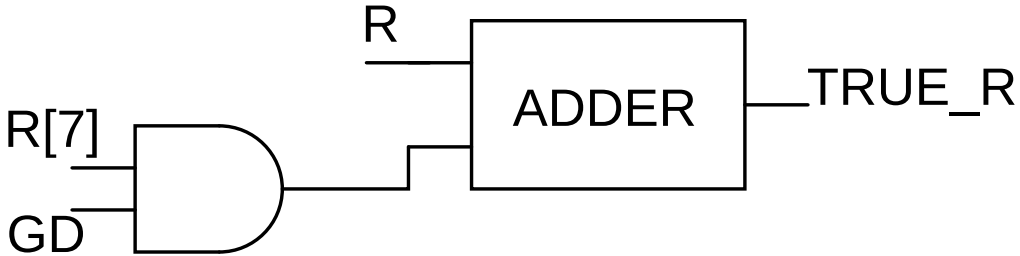


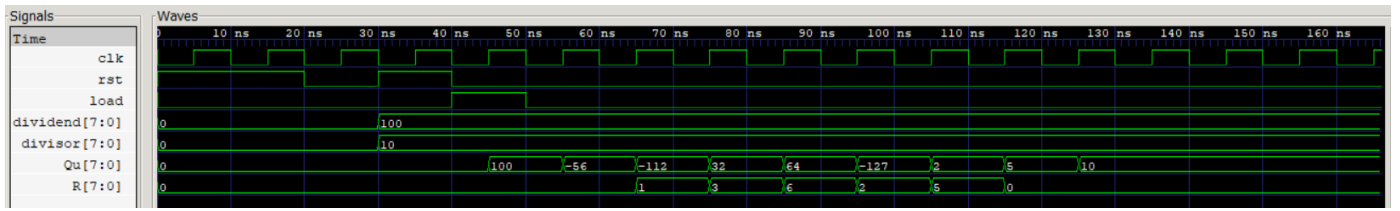
Figure 7: Correction Module Schematic for Resolving Lingering Negative Remainders

A combinatorial safety net resolving any negative remainder left at cycle 8:

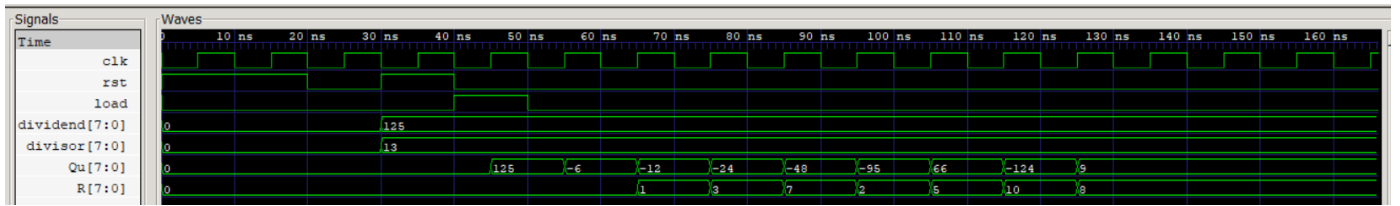
- **Sign Masking (GD):** Uses the final sign bit ($R[7]$) across an AND gate array to generate the Gated Divisor (GD), adding it to the remainder only if negative.
- **9-Bit Padding:** Expands operands to 9 bits temporarily during the check to prevent sign-bit overflow before outputting the clean 8-bit `True_R`.

2.4 Simulation Results

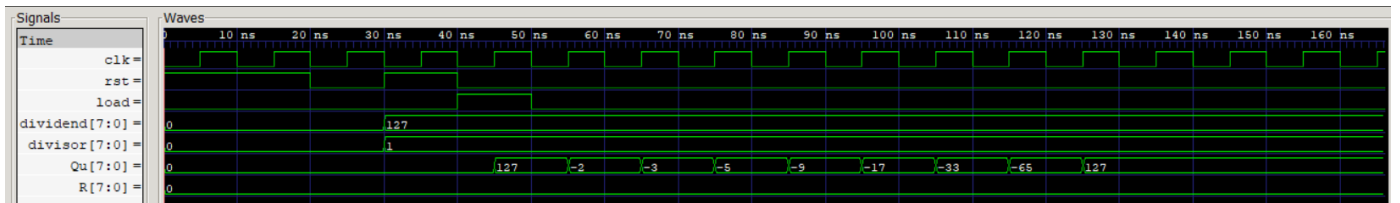
The non-restoring architecture's waveform test results are displayed below.



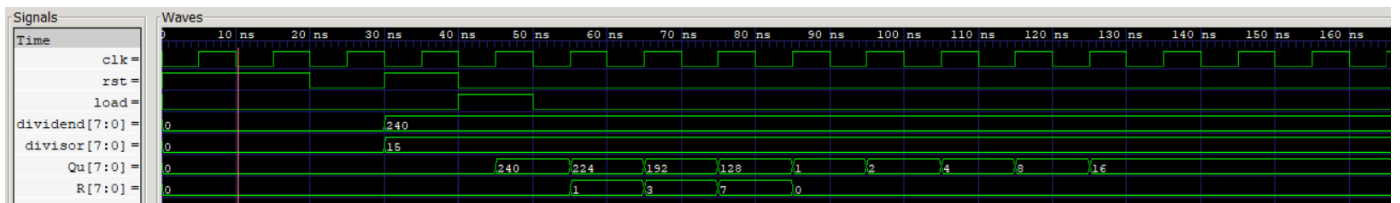
(a) Test Case 1: 100 / 10 (Q=10, R=0)



(b) Test Case 2: 125 / 13 (Q=9, R=8)



(c) Test Case 3: 127 / 1 (Q=127, R=0)



(d) Test Case 4: 240 / 15 (Q=16, R=0)

Figure 8: Timing Simulation Results for the Non-Restoring Divider.

2.5 Detailed Cycle-by-Cycle Trace ($100 \div 10$)

The signed 8-bit shift register renders intermediate negative values as two's complement integers in the waveform viewer. Below is the exact trace corresponding to waveform shown above in simulation result for **est Case 1**:

- **Initial Setup:** Divisor = 10, Initial Qu = 100, Remainder = 0.
- **Cycle 1:**
 - **Action and Bus State:** Shifting left yields 11001000, displayed as signed **-56**.
 - **Math Operation:** $0 - 10 = -10$ (Negative $\rightarrow$ next cycle switches to **addition**). Remainder $R = 0$.
- **Cycle 2:**
 - **Action and Bus State:** Shift yields 10010000, displayed as **-112**.
 - **Math Operation:** ALU adds shifted partial remainder to 10. Yields negative intermediate value; Remainder R updates to **1**.
- **Cycle 3:**
 - **Action and Bus State:** Shift yields 00100000, displayed as positive **32**.
 - **Math Operation:** ALU performs addition. Remainder R updates to **3**.
- **Cycle 4:**
 - **Action and Bus State:** Shift yields 01000000, displayed as **64**.
 - **Math Operation:** Processed through datapath loop. Remainder R updates to **6**.
- **Cycle 5:**
 - **Action and Bus State:** Shift yields 10000001, displayed as **-127**.
 - **Math Operation:** Processed via ALU rules. Remainder R updates to **2**.
- **Cycle 6:**
 - **Action and Bus State:** Shift yields 00000010, displayed as positive **2**.
 - **Math Operation:** Processed through datapath. Remainder R updates to **5**.
- **Cycle 7:**
 - **Action and Bus State:** Shift yields 00000101, displayed as **5**.
 - **Math Operation:** Final bits processed. Remainder R updates to **0**.
- **Cycle 8 (Final):**
 - **Action and Bus State:** Datapath freezes. Qu locks onto final quotient **10**, and Remainder reads **0**.
 - **Correction Module Role:** Final remainder was already positive ($R[7] = 0$), so the CORR block's Gated Divisor (GD) evaluated to 0. The adder performed $R + 0$, leaving the remainder cleanly at 0 without extra adjustments.

3 SRT Divider

3.1 Algorithm and Working Principle

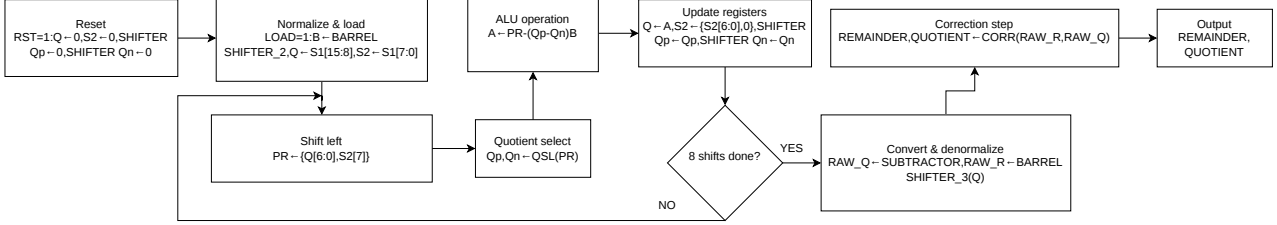


Figure 9: Logical Flowchart of the SRT Division Algorithm

In traditional division algorithms, the hardware is forced to perform an arithmetic operation—either addition or subtraction—on every single clock cycle. This brute-force approach wastes both time and power. The Radix-2 SRT (Sweeney, Robertson, and Tocher) algorithm is a highly optimized alternative that “looks ahead” at the data to skip unnecessary calculations.

The core innovation of the SRT algorithm is the introduction of a redundant digit set: $\{-1, 0, 1\}$. Instead of forcing the quotient bit to be strictly 0 or 1, the hardware estimates the top few bits of the partial remainder to make one of three decisions per cycle:

- **If the remainder is largely Positive:** The quotient digit is +1. The ALU performs a **subtraction**.
- **If the remainder is largely Negative:** The quotient digit is -1. The ALU performs an **addition**.
- **If the remainder is close to Zero:** The quotient digit is 0. The ALU **bypasses math entirely**, and the datapath simply shifts the bits! This “do nothing” state allows the circuit to save significant switching power.

3.2 Data Constraints, Padding, and Normalization

To ensure the hardware logic mathematically resolves without sign-bit corruption, the inputs rely on strict bit-width boundaries and dynamic scaling.

- **Divisor (7-Bit Limit):** The divisor is limited to a maximum decimal value of 127. Its Most Significant Bit (MSB) must remain 0. This successfully isolates the sign bit during Two's Complement ALU arithmetic and Quotient Select Logic (QSL) evaluation.
- **Dividend (8-Bit):** The system safely accepts a full 8-bit unsigned dividend (up to 255).
- **Normalization (Barrel Shifting):** For the QSL estimation to work, the divisor must be scaled to a specific bounded range. `BARREL_SHIFTER_2` shifts the divisor left until its MSB is 1, recording the number of shifts as an offset.
- **The 8-Zero Padding:** To maintain strict mathematical proportion, the dividend must be shifted left by the identical offset. `BARREL_SHIFTER_1` pads the 8-bit dividend with 8 leading zeros to create a 16-bit bus. When the dividend shifts left, its bits slide seamlessly into the upper byte without being lost.

3.3 Hardware Architecture and Internal Components

The implemented Verilog architecture maps this theory into a highly efficient gate-level datapath. Based on the schematics, the primary structures operate as follows:

3.3.1 1. The Main Datapath and Shifting

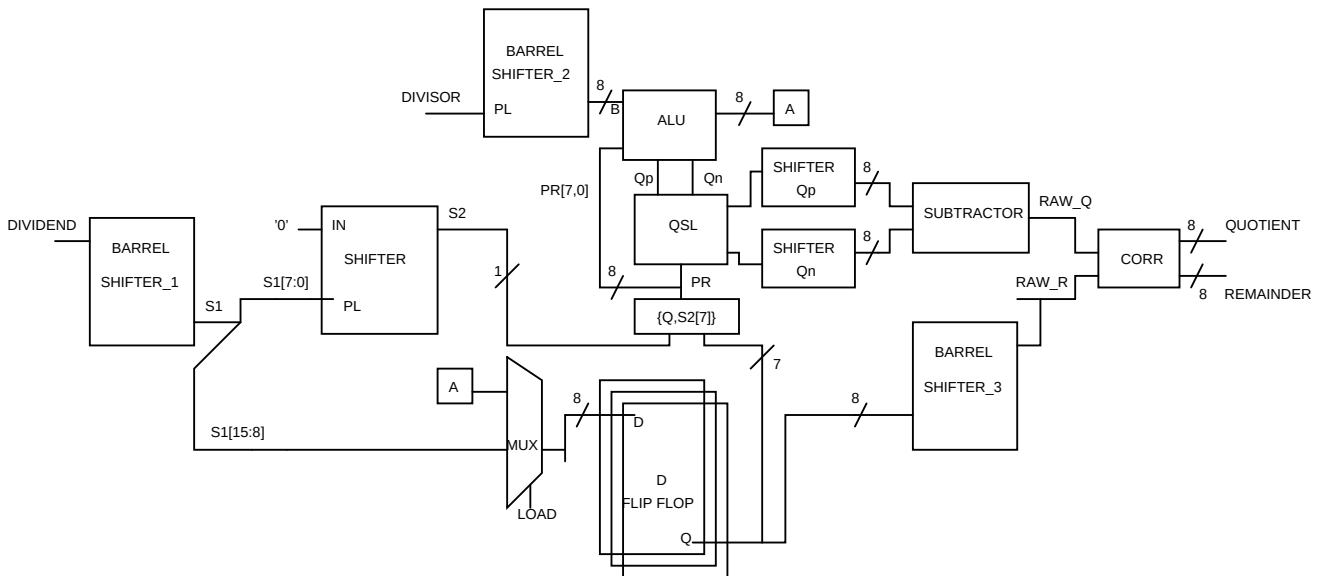


Figure 10: Main Circuit Schematic of the SRT Datapath

Unlike traditional dividers that shift after the math, the proposed architecture performs a combinatorial shift before the ALU. The 16-bit padded dividend is split: the lower bits

enter a dynamic **SHIFTER**, and the upper bits enter a **D FLIP FLOP** array. These wires dynamically concatenate to form the Partial Remainder (**PR**), passing directly into the logic blocks.

3.3.2 2. Quotient Select Logic (QSL)

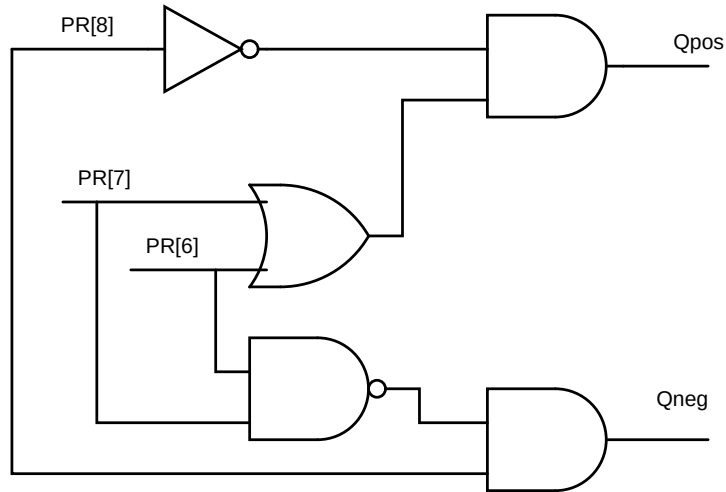


Figure 11: Quotient Select Logic (QSL) Schematic

The QSL module acts as the brain of the SRT divider. It inspects only the top 3 bits of the padded partial remainder: **PR[8]** (acting as the sign bit through a NOT gate) and **PR[7]**, **PR[6]** (the magnitude bits).

- If the sign is positive and the magnitude is large enough, it asserts **Qpos** (+1).
- If the sign is negative and the magnitude is sufficiently negative, it asserts **Qneg** (−1).
- If the magnitude is small, neither wire asserts, yielding a 0 digit.

3.3.3 3. The 3-Way ALU

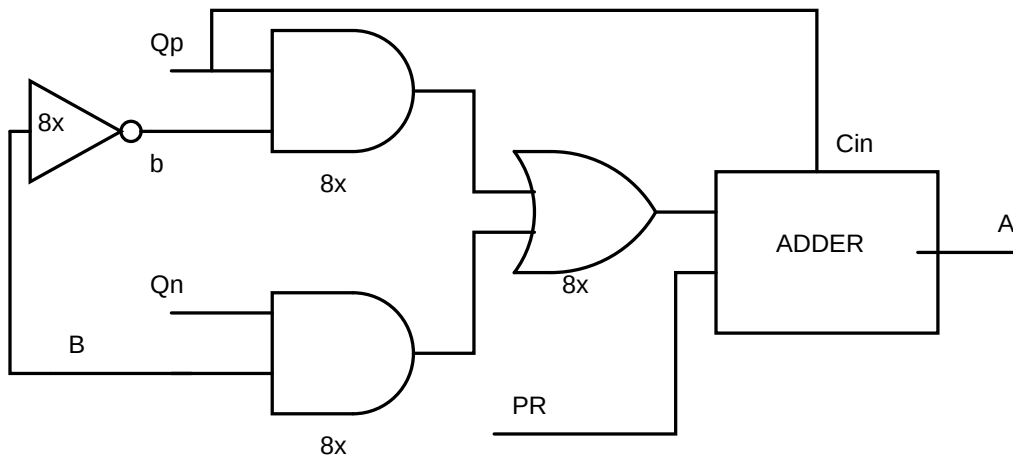


Figure 12: ALU Schematic Featuring 3-Way Bypassing Logic

The ALU uses pure AND/OR combinatorial gating to act as a dynamic 3-way switch driven by the QSL.

- **Subtraction:** If Q_{pos} (Q_p) is high, the upper AND gate passes the inverted divisor ($\sim B$), and the Carry-In (C_{in}) is set to 1. The ADDER executes Two's Complement subtraction.
- **Addition:** If Q_{neg} (Q_n) is high, the lower AND gate passes the normal divisor (B), and C_{in} is set to 0, executing addition.
- **Bypass:** If both Q_{pos} and Q_{neg} are low, the OR gates forcefully mask the divisor to all zeros. The ADDER simply computes $PR + 0$, passing the data through unchanged as output A .

3.3.4 4. Quotient Recombination

Standard binary registers cannot natively store a -1 . Thus, the parallel registers **SHIFTER** Q_p and **SHIFTER** Q_n capture the positive and negative decisions, respectively. A dedicated **SUBTRACTOR** recombines them on the fly using Two's Complement arithmetic to generate the intermediate RAW_Q . **BARREL SHIFTER** $_3$ denormalizes the remainder to produce RAW_R .

3.3.5 5. The Correction Module (CORR)

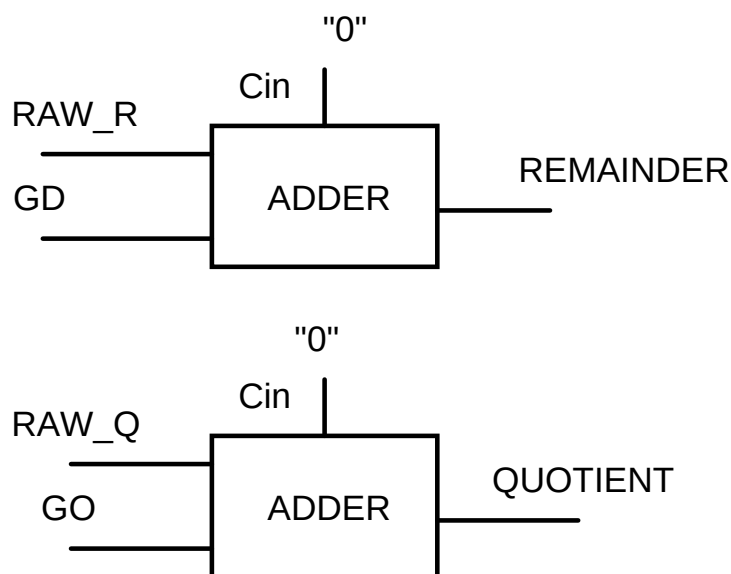


Figure 13: Main Correction Module Schematic

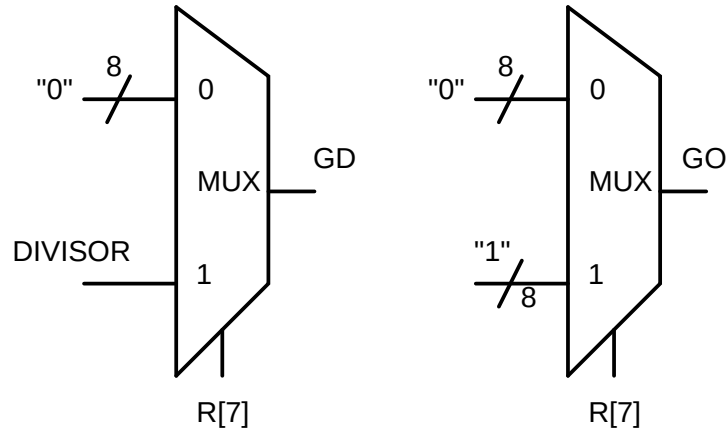


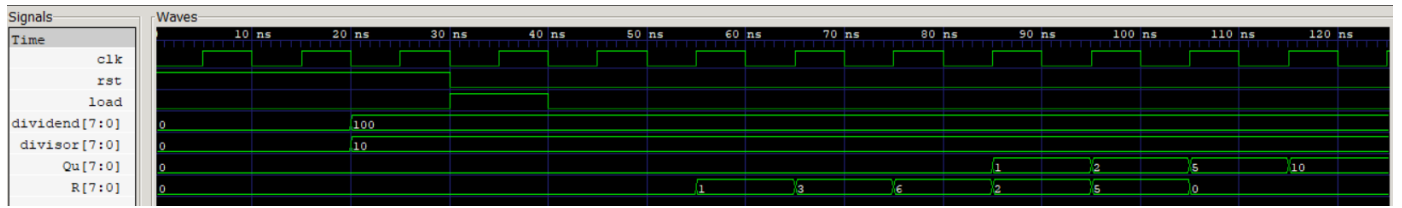
Figure 14: Gated Offset and Gated Divisor Logic within the Correction Module

Because the final cycle can occasionally yield a negative remainder, the CORR module serves as a final safety net. It uses the sign bit of the raw remainder ($R[7]$) to drive a set of multiplexers.

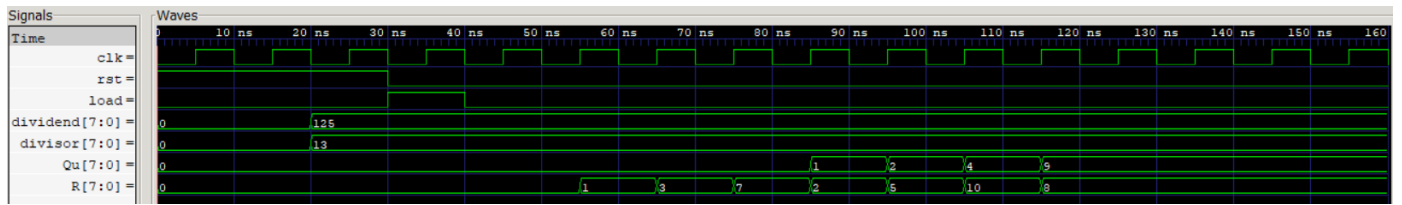
- If the remainder is positive ($R[7] = 0$), the multiplexers output zeros for the Gated Divisor (GD) and Gated Offset (GO). The final adders compute $RAW_R + 0$ and $RAW_Q + 0$, passing the results unchanged.
- If the remainder is negative ($R[7] = 1$), the multiplexers output $GD = DIVISOR$ and $GO = 1$. The adders compute $RAW_R + GD$ to restore the remainder, and $RAW_Q + GO$ to correct the final quotient mathematically.

3.4 Simulation Results

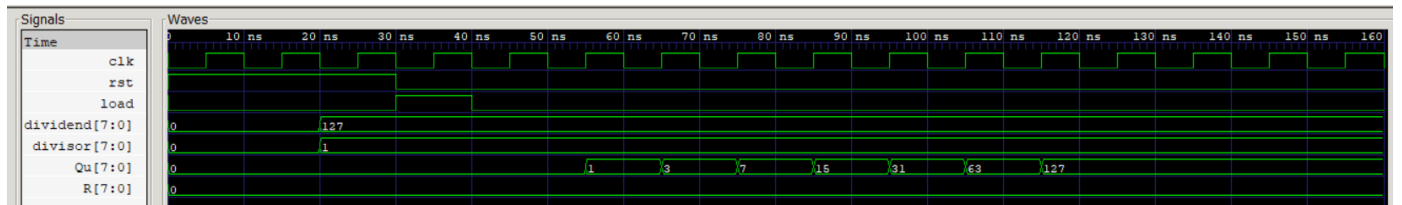
The SRT architecture's waveform test results are displayed below.



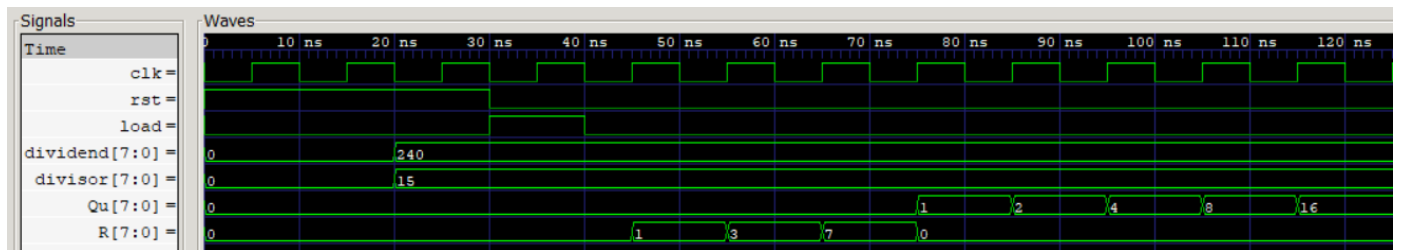
(a) Test Case 1: 100 / 10 (Q=10, R=0)



(b) Test Case 2: 125 / 13 (Q=9, R=8)



(c) Test Case 3: 127 / 1 (Q=127, R=0)



(d) Test Case 4: 240 / 15 (Q=16, R=0)

Figure 15: Timing Simulation Results for the SRT Divider.

3.5 Detailed Cycle-by-Cycle Trace ($100 \div 10$)

This trace explicitly follows the binary data flowing through the combinatorial shifted wires (PR) and matches the theoretical bounds of the architecture and the waveform 2.1.png.

- **Initial Load & Normalization:**

- Divisor 10 (00001010) shifts left by 3 to normalize. Normalized Divisor $B = 80$ (01010000).
- Padded Dividend (00000000_01100100) shifts left by 3 to become 00000011_00100000.
- The upper byte 3 loads into the main tracking register.

- **Cycles 1 to 3:**

- The combinatorial shift evaluates to 6, then 12, then 25.
- The QSL evaluates the top bits (PR[7] and PR[6]) as 0. Magnitude is too small.
- The QSL outputs Qpos=0, Qneg=0. The ALU bypasses math entirely.
- The raw internal quotient remains 0.

- **Cycle 4:**

- The combinatorial shift pushes the internal remainder to 50.
- QSL continues to see top bits as small and bypasses arithmetic.

- **Cycle 5:**

- The combinatorial shift evaluates to 100 (01100100).
- The QSL inspects PR[7]=1, PR[6]=1. The magnitude is finally large enough. It asserts Qpos=1 (Subtract).
- ALU subtracts: $100 - 80 = 20$.
- Recombined internal quotient updates to 1.

- **Cycle 6:**

- The shift pushes the remainder to 40 (00101000).
- QSL sees PR[7]=0, PR[6]=0. Selects 0 (Bypass).
- ALU bypasses math. Internal quotient doubles to 2.

- **Cycle 7:**

- The shift pushes the remainder to 80 (01010000).
- QSL sees PR[7]=0, PR[6]=1. It asserts Qpos=1 (Subtract).
- ALU subtracts: $80 - 80 = 0$.
- Internal quotient updates to 5.

- **Cycle 8 (Final Evaluation):**

- The datapath freezes. The remainder sits at 0. The shifting pushes the quotient to 10.
- **Correction Step:** The CORR module checks $R[7]$. Since the raw remainder is 0 (Positive), the multiplexers force GD and GO to 0.
- **Final Output:** QUOTIENT locks cleanly at 10, and REMAINDER locks perfectly at 0.

4 Newton-Raphson Divider

4.1 Algorithm and Mathematical Foundation

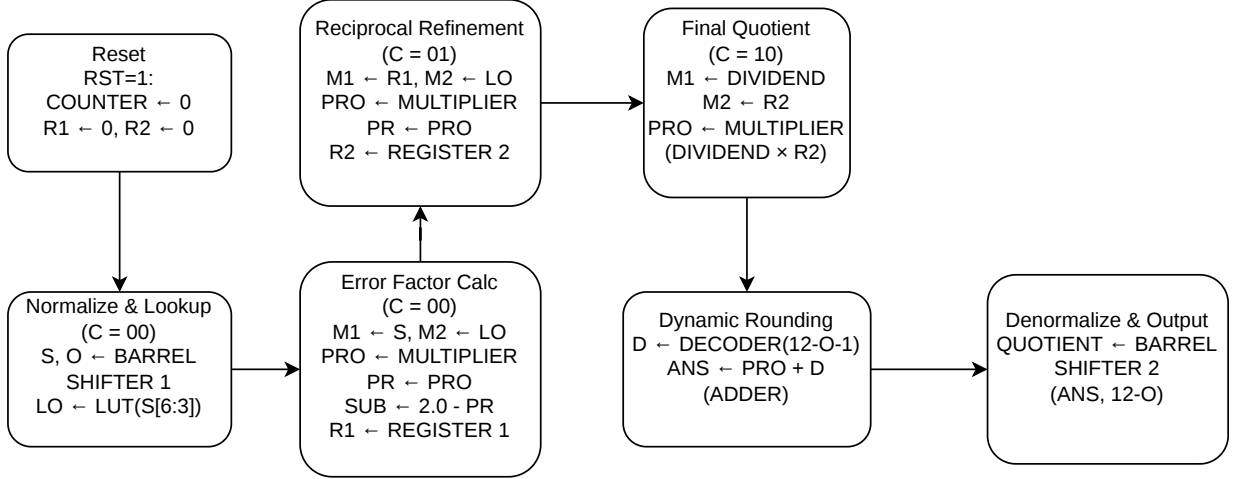


Figure 16: Logical Flowchart of the Newton-Raphson Division Algorithm

Unlike digit-recurrence algorithms (such as Restoring and SRT) that calculate the quotient bit-by-bit using subtraction, the Newton-Raphson division algorithm is a multiplicative approach. It transforms the problem of division into a series of multiplications and additions.

To compute $Q = N \div D$ (where N is the dividend and D is the divisor), the algorithm first approximates the reciprocal of the divisor ($1/D$) and iteratively refines it. Once a highly accurate reciprocal is found, it is simply multiplied by the dividend to produce the final quotient:

$$Q = N \times \frac{1}{D}$$

Because this method relies heavily on multiplication, it features **quadratic convergence**, meaning the number of accurate bits roughly doubles with every iteration.

To find the reciprocal of D , we define a function $f(X)$ that has a root at $X = 1/D$:

$$f(X) = \frac{1}{X} - D = 0$$

Using the standard Newton-Raphson root-finding formula ($X_{i+1} = X_i - f(X_i)/f'(X_i)$) and substituting our function and its derivative, we arrive at the core iteration equation mapped into the hardware:

$$X_{i+1} = X_i(2 - D \cdot X_i)$$

Where:

- X_i is the current approximation of the reciprocal.
- X_{i+1} is the next, more accurate approximation.
- D is the normalized divisor.

4.2 Data Formatting and The Look-Up Table (LUT)

In digital hardware, registers only hold standard binary bits; there is no physical decimal point. The architecture manages complex fractional mathematics by utilizing rigid fixed-point binary formats.

4.2.1 Why Q2.6 Format?

The Look-Up Table (LUT) provides the initial reciprocal guess (X_0). It utilizes the **Q2.6 format**, meaning 2 bits are reserved for the integer portion, and 6 bits are reserved for the fractional decimals. At the start of the algorithm, the divisor is normalized to a fraction between 0.5 and 0.999. Dividing 1 by any number in this range mathematically yields a result between 1.0 and 2.0. To represent a maximum value of 2.0, the hardware strictly requires at least two integer bits (10 in binary). The remaining 6 bits in the 8-bit register are dedicated to the fractional part to preserve maximum accuracy.

4.2.2 Mathematical Derivation and "Bumping"

The 4-bit LUT index (`D_normalized[6:3]`) divides the input range into 16 discrete intervals. The baseline values were calculated by finding the ideal floating-point reciprocal at the midpoint of each interval and scaling it by 2^6 for Q2.6 format. However, because intermediate hardware operations truncate the lower bits to save area, the finite word-length introduces a negative rounding error. To counteract this, an **empirical pre-compensation bias** (the "bumped" values) was added to the baseline reciprocals. This static bias acts as an artificial ceiling, preventing values from dropping too low and actively shielding edge cases like division by 1, 2, or prime numbers.

4.2.3 Example: Deriving the LUT Value for Divisor 10

To understand how fixed-point math works in practice, let's look at the division of $100 \div 10$.

- **Normalization (The Imaginary Decimal):** The divisor 10 (`00001010_2`) is shifted left by 4 positions to become `10100000_2`. If you read this as a normal whole number, it equals 160. However, the hardware places an imaginary decimal point at the very front of the register to treat it as a fraction (Q0.8 format). To find its real-world decimal value, you take the raw integer (160) and divide it by the maximum possible value of an 8-bit box (which is 256). $160 \div 256 = 0.625$. The hardware acts as if the divisor is exactly 0.625.
- **Ideal Reciprocal:** Instead of calculating $1 \div 160$, the math requires $1 \div 0.625$. The exact mathematical reciprocal is 1.6.
- **Q2.6 Conversion:** To store the number 1.6 inside our Q2.6 LUT, we multiply it by 2^6 (64). $1.6 \times 64 = 102.4$. Truncating the decimals gives us a baseline binary value of 102.
- **The "Bump":** The LUT index for this normalized divisor is extracted from bits `[6:3]`, which is `0100_2` (Index 4). Instead of storing the exact baseline of 102, the hardware stores a "bumped" value of 103. This deliberate +1 pre-compensation shields the subsequent multiplier truncation, ensuring the intermediate values do not floor down during the iterative calculation.

4.2.4 The 1-Iteration Architectural Trade-Off

For a 4-bit initial LUT guess, a single iteration of the Newton-Raphson formula doubles the precision to roughly 8 bits. While technically sufficient for an 8-bit divider, hardware truncation means the final quotient may occasionally exhibit an error of ± 1 in the Least Significant Bit (LSB). Achieving 100% bit-perfect accuracy across all combinations would require either a massive 12-bit internal datapath or executing a second full iteration. Limiting the hardware to a single iteration and accepting a ± 1 LSB tolerance is a deliberate architectural trade-off, prioritizing extremely high-speed operation and low resource utilization over absolute integer perfection.

4.3 Hardware Architecture and Circuit Flow

The complete hardware execution is orchestrated by the top-level main module, which routes data through the datapath over three distinct clock cycles using a 2-bit counter.

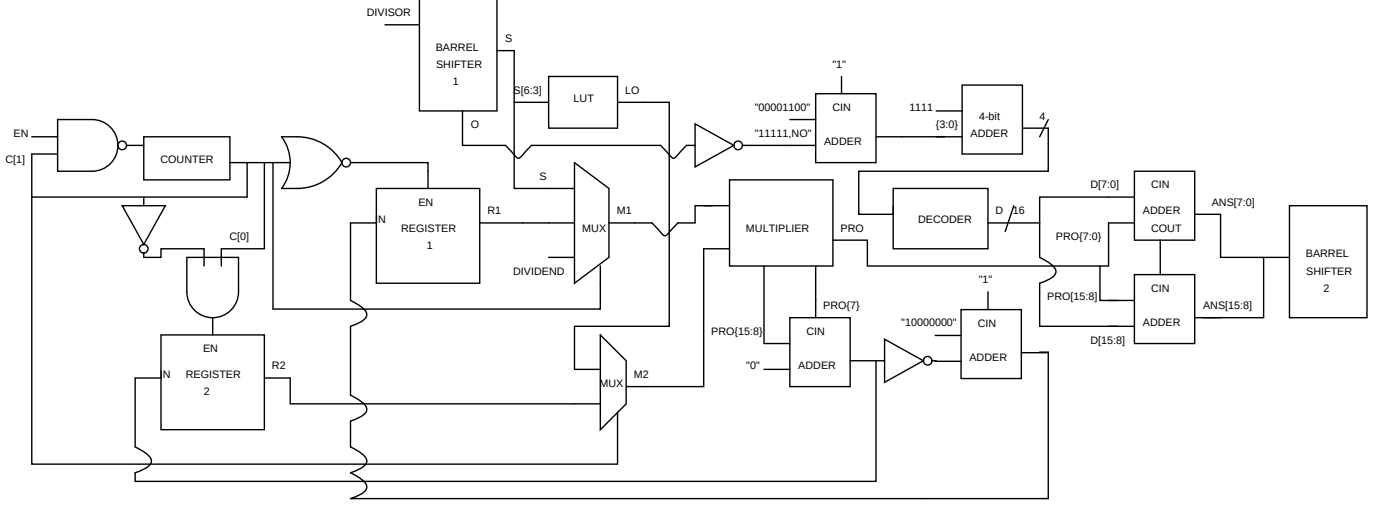


Figure 17: Top-Level Main Circuit Architecture (**MAIN.ps**)

As depicted in Figure 17, the physical flow of data is governed by structural multiplexing:

1. **Normalization & Seed Loading:** The divisor enters **BARREL SHIFTER 1**, which dynamically shifts the bits left until the MSB is 1. The shift amount is extracted as offset o . The top fractional bits are fed into the LUT to immediately generate the seed X_0 .
2. **Cycle 0 (Error Calculation):** Multiplexers **M1** and **M2** route the normalized divisor and X_0 into the shared multiplier. The product immediately flows into the Two's Complement subtractor to evaluate $2 - D \cdot X_0$. The resulting error term is locked into **REGISTER 1**.
3. **Cycle 1 (Reciprocal Refinement):** The counter increments. The multiplexers now select X_0 and the contents of **REGISTER 1**. The multiplier computes the refined reciprocal X_1 , which is captured by **REGISTER 2**.
4. **Cycle 2 (Quotient Generation):** The multiplexers select the original Dividend and the refined reciprocal X_1 from **REGISTER 2**. The final unscaled product passes through the dynamic +0.5 rounding adder and enters **BARREL SHIFTER 2**, which denormalizes the data to produce the true final quotient.

4.4 High-Speed Multiplier Architecture

The Newton-Raphson algorithm requires two multiplications per iteration. A standard ripple-carry multiplier would create a massive bottleneck. To overcome this, the architecture utilizes a parallel multiplier tree, operating entirely independently of clocks.

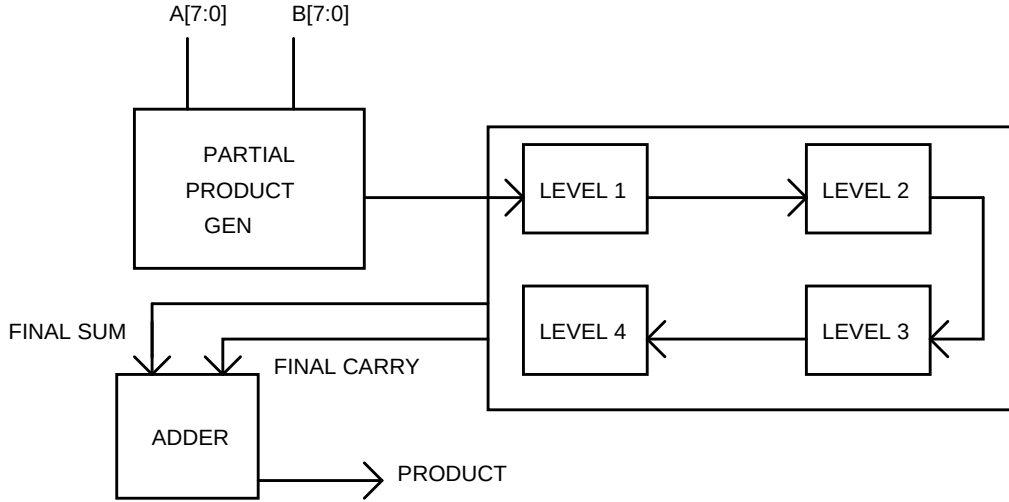


Figure 18: Parallel Multiplier Architecture Featuring 4-Level Compression (`multi.ps`)

4.4.1 Phase 1: Partial Product Generation (The 4-Bit Example)

To understand data flow, we can trace a 4-bit multiplication of 13 (1101_2) by 11 (1011_2). The hardware uses AND gates to generate rows based on the multiplier bits. If the bit is 1, it copies the multiplicand; if 0, it outputs zeros:

Row 0 ($B[0] = 1$) :	1 1 0 1
Row 1 ($B[1] = 1$) :	1 1 0 1
Row 2 ($B[2] = 0$) :	0 0 0 0
Row 3 ($B[3] = 1$) :	1 1 0 1

4.4.2 Phase 2: The Column Matrix and Compression

Instead of adding these rows sequentially, the hardware rearranges them vertically into columns (c_6 down to c_0). These columns enter the parallel compression tree to squash the tall vertical stacks down to a maximum height of two bits.

- **Col 0:** Contains 1. $\rightarrow$ Sum = 1.
- **Col 1:** Contains $0 + 1$. $\rightarrow$ Sum = 1.
- **Col 2:** Contains $1 + 0 + 0$. $\rightarrow$ Sum = 1.
- **Col 3:** Contains $1 + 1 + 0 + 1 = 3_{10}$ (11_2). $\rightarrow$ Sum = 1, Carry = 1 into Col 4.
- **Col 4:** Contains $1 + 0 + 0 + \text{Carry}(1) = 2_{10}$ (10_2). $\rightarrow$ Sum = 0, Carry = 1 into Col 5.
- **Col 5:** Contains $0 + 1 + \text{Carry}(1) = 2_{10}$ (10_2). $\rightarrow$ Sum = 0, Carry = 1 into Col 6.
- **Col 6:** Contains $1 + \text{Carry}(1) = 2_{10}$ (10_2). $\rightarrow$ Sum = 0, Carry = 1 into Col 7.
- **Col 7:** Final Carry(1). $\rightarrow$ Sum = 1.

4.4.3 Phase 3: Final Addition

In the full 8x8 architecture, a 4-level tree reduces the entire matrix concurrently into a `final_sum` vector and a `final_carry` vector. A standard fast adder merges them. Reading our 4-bit example sums yields `10001111_2` (143_{10}), instantly confirming 13×11 .

4.5 Top-Level Arithmetic and Scaling Logic

4.5.1 A. The Two's Complement Subtractor ($2 - D \cdot X_0$)

Digital hardware does not have a dedicated "subtract" gate. Because the output is in Q2.6 format, the decimal value 2.0 is represented as binary `10000000` (128). To perform the subtraction, the circuit takes the 8-bit product, passes it through NOT gates, and feeds it into an adder along with the constant 128 and a Carry-In of 1. This evaluates $128 + (\sim\text{Product} + 1)$, perfectly executing Two's Complement subtraction without a dedicated sub-module.

4.5.2 B. The Mathematical Derivation of the $(12 - k)$ Denormalization Shift

At the beginning of the algorithm, the divisor was shifted left by k bits to normalize it. This artificial scaling must be reversed at the end. The required right-shift amount is exactly $12 - k$ due to how the hardware manages the scale factor across the datapath cycles:

1. **Initial Scale:** The normalized divisor is $D \times 2^k$. The Q2.6 LUT guess (X_0) is implicitly scaled by 2^{14} relative to this.
2. **Cycle 1 (Error Term):** The multiplier computes $D_{norm} \times X_0$. The hardware chops off the lower 8 bits (dividing by 2^8). The current scale is $2^{14} \div 2^8 = 2^6$. This aligns perfectly with the constant 128 (which is 2×2^6).
3. **Cycle 2 (Refined Reciprocal):** The multiplier computes $X_1 = X_0 \times \text{Error}$. Multiplying their respective scales ($2^{14} \times 2^6$) yields a temporary scale of 2^{20} . The hardware chops the lower 8 bits again. The refined reciprocal X_1 is now inherently scaled up by a factor of 2^{12} .
4. **Cycle 3 (Final Adjustment):** The final product is $N \times X_1$. Because the reciprocal is scaled by 2^{12} and the divisor was originally multiplied by 2^k , the internal product evaluates to: $\text{Product} = \frac{N}{D} \times 2^{12-k}$.

To extract the true whole-number quotient, the hardware calculates $(12 - k)$ and right-shifts the product by this exact amount.

4.5.3 C. Structural Dynamic Rounding (+0.5 LSB) via Decoder

When hardware shifts a binary number to the right to denormalize it, it naturally chops off the lower bits, which mathematically acts as a floor function (rounding down). To achieve the most mathematically accurate integer quotient, the hardware must round to the nearest whole number by adding 0.5 (a binary 1) precisely one position to the right of the active cutoff point before shifting.

Because the final right-shift amount $(12 - k)$ changes dynamically depending on the input, the cutoff point also moves. The hardware calculates this using a dedicated structural routing block:

- **Identifying the Target:** The circuit subtracts 1 from the $(12 - k)$ shift amount (by adding the constant 1111), pinpointing the exact bit position just below the cutoff line.
- **The Decoder Logic:** This target is fed into a 4-to-16 Decoder. The decoder is a pure combinatorial block built from 4 NOT gates and 16 four-input AND gates. It maps the 4-bit target to a 16-bit "one-hot" mask, where only the exact targeted bit is activated as a 1. By using standard logic gates, the hardware achieves this dynamic bit-shifting without relying on slow, behavioral << software operators.
- **The Injection:** This generates a one-hot vector that flawlessly injects a single 1 into the exact binary cutoff position before truncation, executing perfect structural rounding.

4.6 Detailed Cycle-by-Cycle Trace ($100 \div 10$)

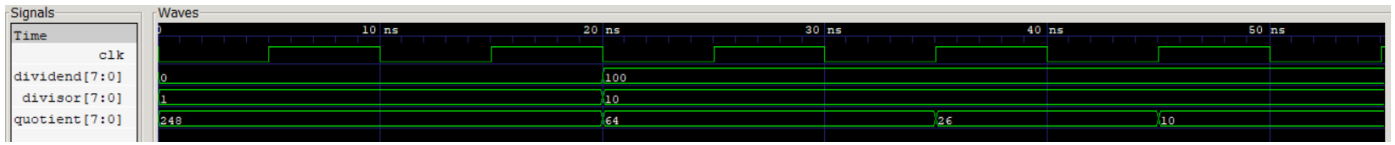
The hardware executes the division of 100 by 10 across three clock cycles.

- **Stage 1: Normalization & LUT Lookup**
 - **Inputs:** Dividend $N = 100$ (01100100_2), Divisor $D = 10$ (00001010_2).
 - **Normalization:** D is shifted left by 4 positions ($k = 4$) so the MSB hits bit 7. $D_{norm} = 160$ (10100000_2).
 - **LUT Guess:** The LUT takes bits [6:3] (0100_2 or 4). At index 4, the Q2.6 pre-compensation value is $X_0 = 103$.
- **Cycle 0: Error Calculation ($2 - D_{norm} \cdot X_0$)**
 - **Multiplication:** $D_{norm} \times X_0 \rightarrow 160 \times 103 = 16480$ (01000000_01100000_2).
 - **Chopping:** The lower 8 bits are chopped (extracting `pro[15:8]`). $16480 \div 2^8 = 64$.
 - **Subtraction:** Two's Complement subtraction: $128 + (\sim 64) + 1 = 64$. The Error register stores 64.
- **Cycle 1: Refined Reciprocal ($X_1 = X_0 \cdot \text{Error}$)**
 - **Multiplication:** $X_0 \times \text{Error} \rightarrow 103 \times 64 = 6592$ (00011001_11000000_2).
 - **Chopping & Carry:** The upper 8 bits (`pro[15:8]`) evaluate to 25. Because bit 7 is a 1, the internal adder carries this bit over, rounding the chopped value up to 26. $X_1 = 26$.
- **Cycle 2: Final Multiplication and Scaling**
 - **Multiplication:** $N \times X_1 \rightarrow 100 \times 26 = 2600$.
 - **Denormalization Shift:** The offset is $12 - k \rightarrow 12 - 4 = 8$.

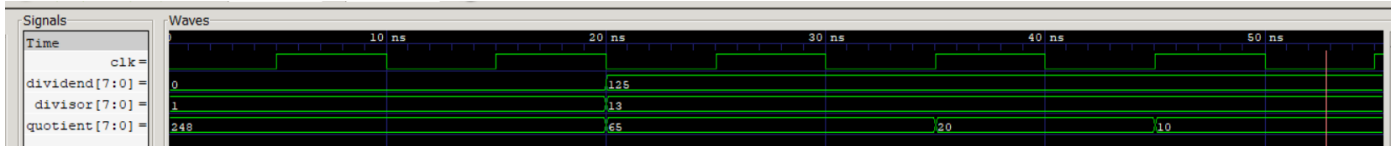
- **Dynamic Rounding:** To add $+0.5$, the decoder targets position $8 - 1 = 7$. Adding 2^7 (128) yields $2600 + 128 = 2728$.
- **Final Shift:** The total is right-shifted by 8. $2728 \gg 8 = 10$.
- **Result:** The hardware correctly outputs the final quotient as exactly 10.

4.7 Simulation Results

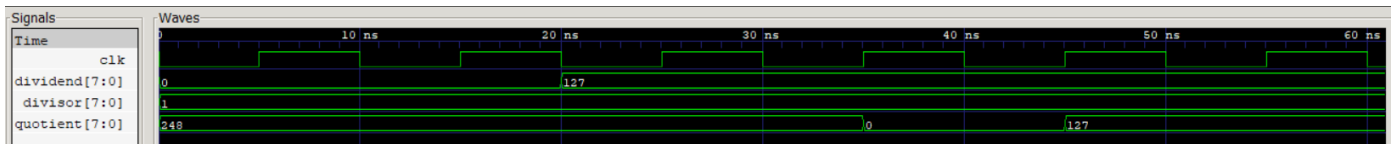
The timing waveforms and operational states of the Newton-Raphson architecture are verified in the simulation results below.



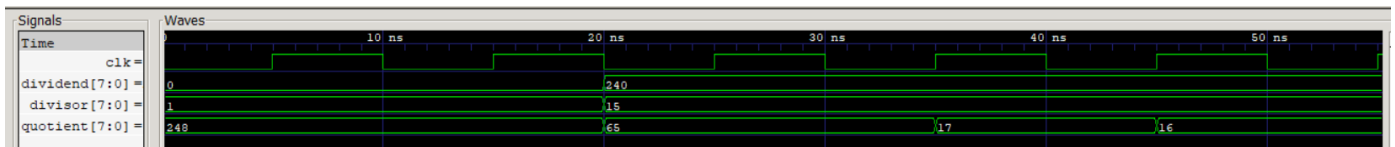
(a) Simulation Result 1



(b) Simulation Result 2



(c) Simulation Result 3



(d) Simulation Result 4

Figure 19: Timing Simulation Results for the Newton-Raphson Divider.

5 Goldschmidt Divider

5.1 Algorithm and Mathematical Foundation

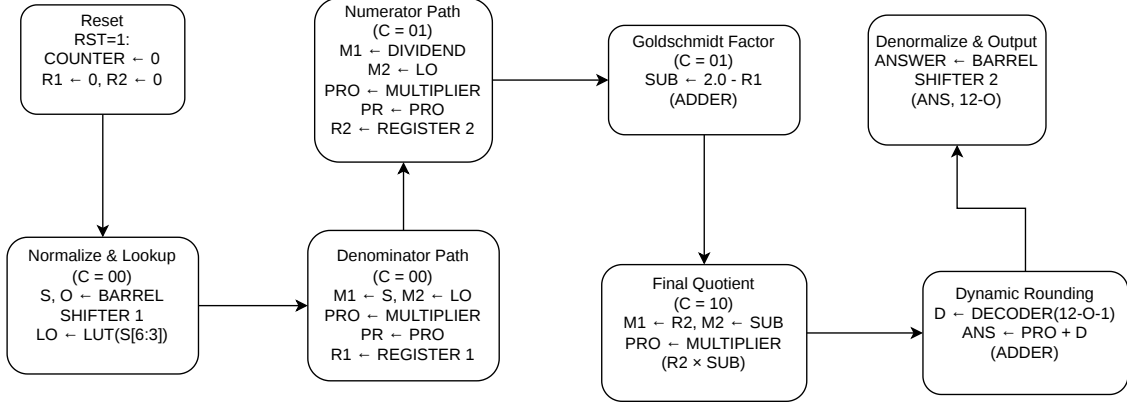


Figure 20: Logical Flowchart of the Goldschmidt Division Algorithm

Unlike the Newton-Raphson method, which strictly focuses on iteratively refining the reciprocal of the divisor before executing a final multiplication, the **Goldschmidt division algorithm** utilizes a simultaneous iterative scaling approach.

To compute $Q = N \div D$ (where N is the dividend and D is the divisor), Goldschmidt's method multiplies both the Numerator (N) and the Denominator (D) by a common convergence factor (F_i) at each step:

$$D_i = D_{i-1} \times F_i$$

$$N_i = N_{i-1} \times F_i$$

The convergence factor is mathematically chosen so that the denominator D_i progressively approaches exactly 1.0. As the denominator is forced toward 1.0, the numerator N_i simultaneously converges directly into the final true quotient Q .

The hardware generates this dynamic scaling factor using Two's Complement logic, calculating it by simply subtracting the currently scaled denominator from the constant 2.0:

$$F_i = 2.0 - D_{i-1}$$

5.2 Sub-Module Reuse and Architectural Approach

The architecture of the Goldschmidt divider is highly optimized. It is constructed using the exact same foundational hardware sub-modules designed for the Newton-Raphson method. The shared library of structural blocks includes:

- **BARREL SHIFTERS:** For the initial divisor normalization and final quotient denormalization.
- **LOOK-UP TABLE (LUT):** The Q2.6 initial reciprocal seed table.

- **PARALLEL MULTIPLIER:** The high-speed 4-level compression multiplier core.
- **ADDERS:** The gate-level Two's Complement adders and subtractors.

While the physical flow of data differs significantly to support simultaneous scaling, the deep internal, gate-level mechanics of these shared components function identically. Therefore, their specific internal behaviors are detailed thoroughly in the previous Newton-Raphson chapter.

5.3 Hardware Architecture and Circuit Flow

Because Goldschmidt scales two separate mathematical paths concurrently, the hardware introduces dedicated state registers (a DENOMINATOR REGISTER and a NUMERATOR REGISTER) alongside structural MULTIPLEXERS to iteratively feed operands back into the shared PARALLEL MULTIPLIER.

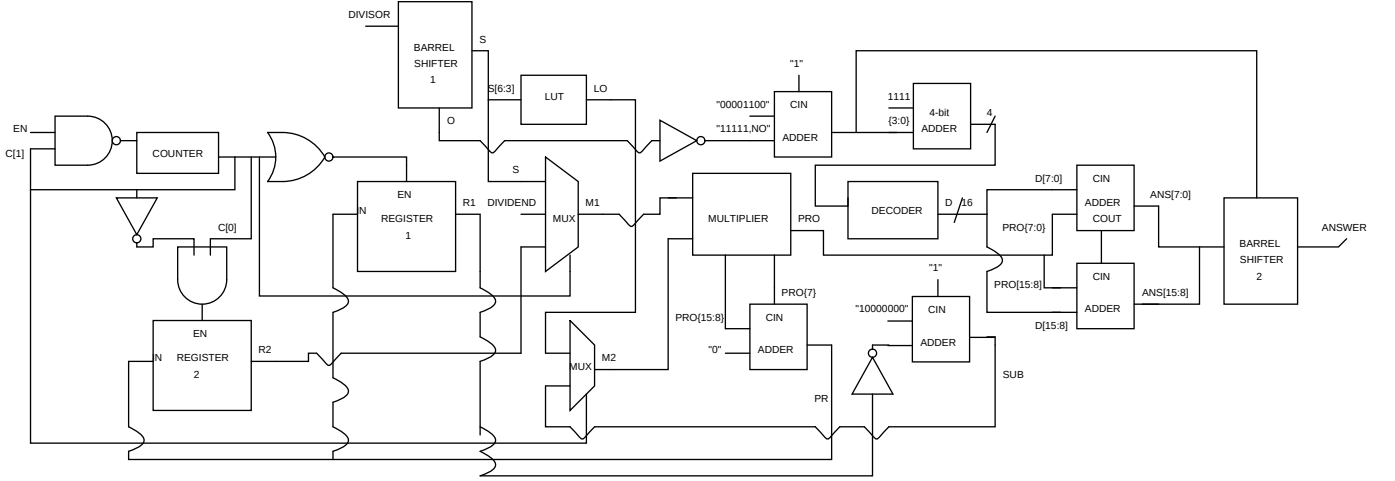


Figure 21: Top-Level Goldschmidt Main Circuit Architecture (`gmain.ps`)

As shown in Figure 21, the complete division is flawlessly orchestrated across three active clock cycles:

1. **Normalization & Seed Loading:** The divisor is shifted left by the BARREL SHIFTER until normalized, setting the shift offset k . The highest fractional bits instantly query the LOOK-UP TABLE to generate the initial convergence factor seed (F_0).
2. **Cycle 0 (Denominator Path):** The MULTIPLEXERS route the normalized divisor (D_{norm}) and the seed (F_0) into the PARALLEL MULTIPLIER. The product is statically rounded and captured by the DENOMINATOR REGISTER, establishing D_1 .
3. **Cycle 1 (Numerator Path):** The MULTIPLEXERS now route the original Dividend (N) and the seed (F_0) into the PARALLEL MULTIPLIER. This product is captured by the NUMERATOR REGISTER, establishing N_1 . Simultaneously, the structural Two's Complement ADDERS compute the next convergence factor: $F_1 = 2.0 - D_1$.

4. **Cycle 2 (Final Quotient):** In a clever hardware optimization, the circuit bypasses calculating D_2 (as it is guaranteed to approach 1.0) and solely calculates the final numerator. The PARALLEL MULTIPLIER processes $N_1 \times F_1$. The result passes through the dynamic rounding logic and is denormalized by the final BARREL SHIFTER to output the final quotient.

5.3.1 Structural Dynamic Rounding (+0.5 LSB) via Decoder

When hardware shifts a binary number to the right to denormalize it, it naturally chops off the lower bits, which mathematically acts as a floor function (rounding down). To achieve the most mathematically accurate integer quotient, the hardware must round to the nearest whole number by adding 0.5 (a binary 1) precisely one position to the right of the active cutoff point before shifting.

Because the final right-shift amount ($12 - k$) changes dynamically depending on the input, the cutoff point also moves. The hardware calculates this using a dedicated structural routing block:

- **Identifying the Target:** A 4-bit ADDER takes the calculated shift amount ($12 - k$) and subtracts 1 by adding the constant 1111, pinpointing the exact bit position just below the cutoff line.
- **The Decoder Logic:** This target is fed into a 4-to-16 DECODER. Though visually abstracted in the top-level diagram, the DECODER is a pure combinatorial block built from 4 NOT gates and 16 four-input AND gates. It maps the 4-bit target to a 16-bit "one-hot" mask, where only the exact targeted bit is activated as a 1. By using standard logic gates, the hardware achieves this target mapping without relying on slow, behavioral $\ll$ software operators.
- **The Injection:** The 16-bit MULTIPLIER product and the 16-bit one-hot mask merge through two chained 8-bit ADDERS, flawlessly injecting the +0.5 rounding value before the final truncation occurs.

5.4 Detailed Cycle-by-Cycle Trace ($100 \div 10$)

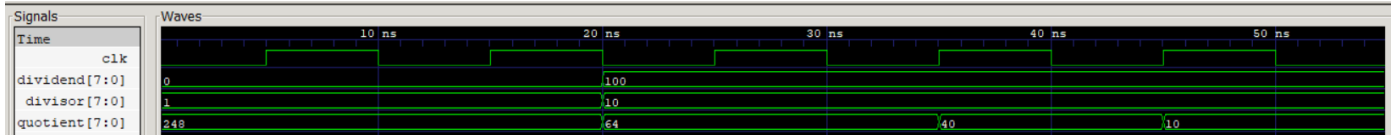
To practically illustrate the simultaneous iterative scaling of the Goldschmidt architecture, we can trace the division of 100 by 10 step-by-step through the fixed-point hardware.

- **Stage 1: Normalization & Initial Factor (F_0)**
 - **Inputs:** Dividend $N = 100$ (01100100_2), Divisor $D = 10$ (00001010_2).
 - **Normalization:** D is shifted left by 4 positions ($k = 4$). $D_{norm} = 160$ (10100000_2). In Q0.8 format, this represents the fraction 0.625.
 - **Initial Factor:** The LOOK-UP TABLE index evaluates to 4. The pre-compensated Q2.6 seed is $F_0 = 103$ (≈ 1.6).
- **Cycle 0: Scaling the Denominator ($D_1 = D_{norm} \cdot F_0$)**
 - **Multiplication:** $160 \times 103 = 16480$ (01000000_01100000_2).
 - **Chopping:** Static rounding extracts the upper 8 bits (pro[15:8]). $16480 \div 256 = 64$.

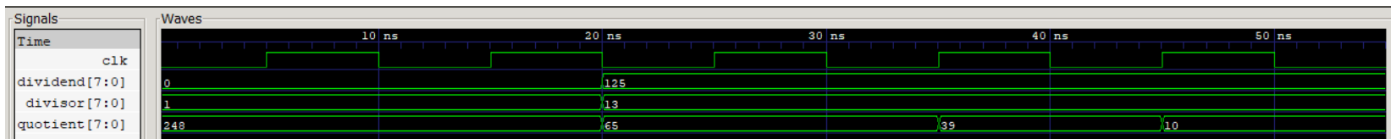
- **Storage:** The scaled denominator $D_1 = 64$ is locked into the DENOMINATOR REGISTER.
- **Cycle 1: Scaling the Numerator ($N_1 = N \cdot F_0$) & Generating Factor F_1**
 - **Multiplication:** The MULTIPLEXERS route N and F_0 . $100 \times 103 = 10300$ (00101000_00111100_2).
 - **Chopping:** Extracting the upper 8 bits provides $10300 \div 256 = 40$.
 - **Storage:** The scaled numerator $N_1 = 40$ is locked into the NUMERATOR REGISTER.
 - **Factor Calculation:** $F_1 = 2.0 - D_1 \rightarrow 128 - 64 = 64$.
- **Cycle 2: Final Product and Denormalization ($N_{final} = N_1 \cdot F_1$)**
 - **Multiplication:** The PARALLEL MULTIPLIER evaluates $N_1 \times F_1 \rightarrow 40 \times 64 = 2560$.
 - **Denormalization Shift:** The final output offset is $12 - k \rightarrow 12 - 4 = 8$.
 - **Dynamic Rounding:** To inject $+0.5$, the DECODER targets position $8 - 1 = 7$. Adding 2^7 (128) yields $2560 + 128 = 2688$.
 - **Final Shift:** The total is structurally right-shifted by 8. $2688 \gg 8 = 10$.
 - **Result:** The numerator path has successfully converged, outputting the exact physical quotient of 10.

5.5 Simulation Results

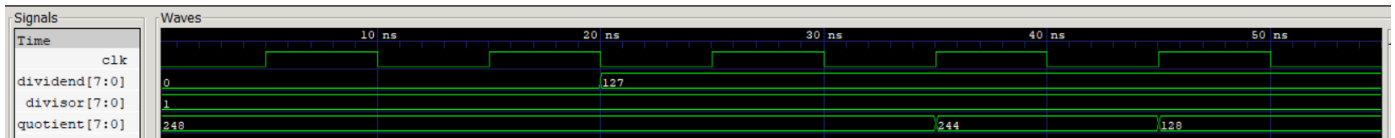
The timing waveforms, concurrent factor scaling, and structural data routing of the Goldschmidt architecture are comprehensively verified in the simulation results presented below.



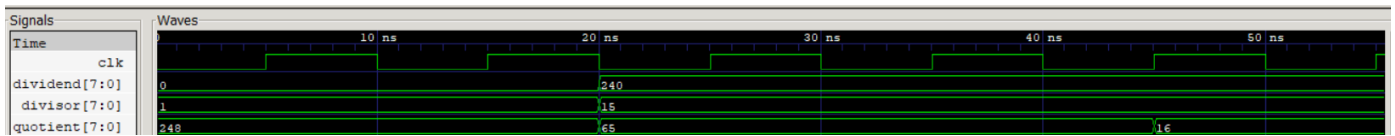
(a) Simulation Result 1



(b) Simulation Result 2



(c) Simulation Result 3



(d) Simulation Result 4

Figure 22: Timing Simulation Results for the Goldschmidt Divider.

6 Comparative Analysis of Divider Architectures

After designing and simulating all five binary dividers for an 8-bit datapath, clear distinctions emerge regarding their speed, area, and overall efficiency. The optimal choice of architecture heavily depends on the required bit-width and hardware constraints.

6.1 Digit-Recurrence (Restoring, Non-Restoring, SRT)

- **Restoring & Non-Restoring:** These are the simplest architectures. They rely on standard adders/subtractors and shift registers. Because they calculate one quotient bit per clock cycle, they require exactly N cycles for an N -bit division. For an 8-bit system, an 8-cycle delay is highly acceptable. The hardware footprint is minimal, making them highly efficient for lightweight microcontrollers.
- **SRT Division:** By utilizing a redundant digit set and predicting quotient bits, SRT skips over consecutive zeros or ones. While slightly more complex than Non-Restoring division, it significantly reduces the number of required additions/subtractions. It strikes an excellent balance, offering a noticeable speedup while maintaining a relatively small hardware footprint.

6.2 Multiplicative (Newton-Raphson, Goldschmidt)

- **High-Speed Convergence:** Both algorithms utilize quadratic convergence, doubling their accuracy with every iteration. They can compute a highly accurate quotient in just 2 to 3 clock cycles.
- **Hardware Overhead:** To achieve this speed, both architectures require massive hardware resources. They demand high-speed parallel multiplier trees (which consume vast amounts of combinatorial logic), Look-Up Tables (LUTs) for initial seed generation, dynamic rounding decoders, and multiple wide registers.

6.3 Conclusion: The 8-Bit Overkill

While Newton-Raphson and Goldschmidt are the standard choice for large-scale 32-bit and 64-bit processors—where saving dozens of clock cycles is critical—they are a severe over-design for a simple 8-bit datapath.

The physical area and power consumed by an 8x8 parallel multiplier tree, coupled with the complex structural rounding logic, vastly outweighs the benefit of saving just 5 clock cycles compared to an SRT or Non-Restoring divider. For an 8-bit system, the digit-recurrence methods provide the optimal engineering solution in terms of power, area, and speed trade-offs. The multiplicative algorithms stand as a demonstration of high-performance architecture rather than a practical necessity for small-scale hardware.